Supermarket Sales Performance Dashboard

An end-to-end analysis of 1,000 supermarket transactions — from exploratory data analysis in Python to structured SQL queries — uncovering sales patterns, customer behavior, and branch performance.

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Dataset at a Glance

1,000

Total Rows

Supermarket sales transactions

17

Total Columns

Numerical, categorical & date/time

0

Missing Values

Clean dataset, no nulls or duplicates

6

Product Lines

Fashion, Food, Electronics & more

Numerical

Unit price, Quantity, Tax, Total, Rating

Categorical

Branch, City, Customer type, Gender, Product line, Payment

Date / Time

Transaction date and time stamps

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Exploratory Data Analysis: Python Workflow

Load Data → **Explore Structure** → **Handle Nulls** → **Standardize Columns**

Python's pandas and seaborn libraries powered the full EDA pipeline — from initial loading through cleaning and column standardization — confirming a complete, duplicate-free dataset ready for analysis.

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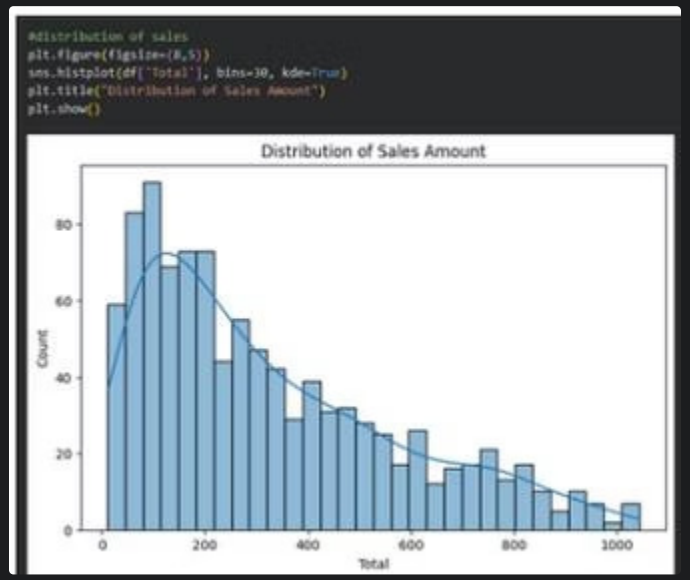
Distribution of Sales Amount



Key Insight

The sales distribution is **right-skewed**, with most transactions concentrated between $10 – $200. The peak falls around $150–$200, while a long tail extends to a maximum of **$1,042.65**.

Mean Total

$322.97 per transaction

Median Total

$253.85 — below the mean, confirming skew

Max Total

$1,042.65 — high-value outlier transactions

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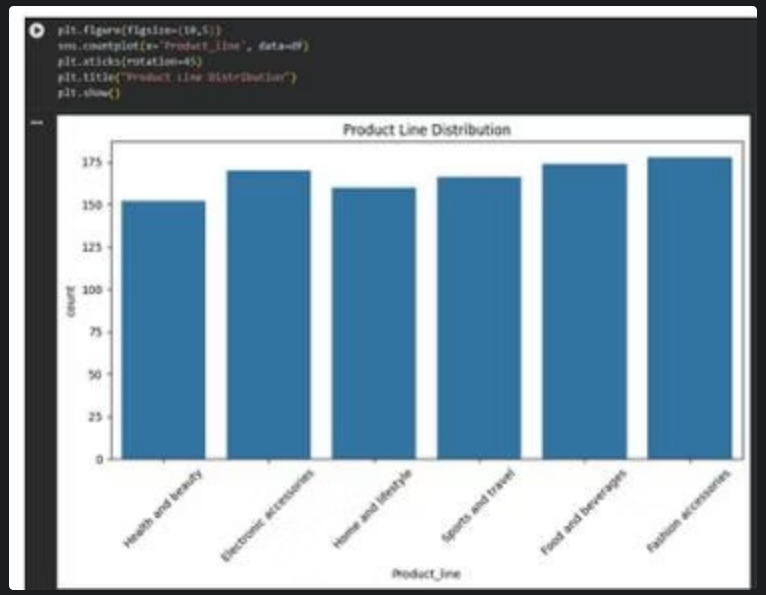
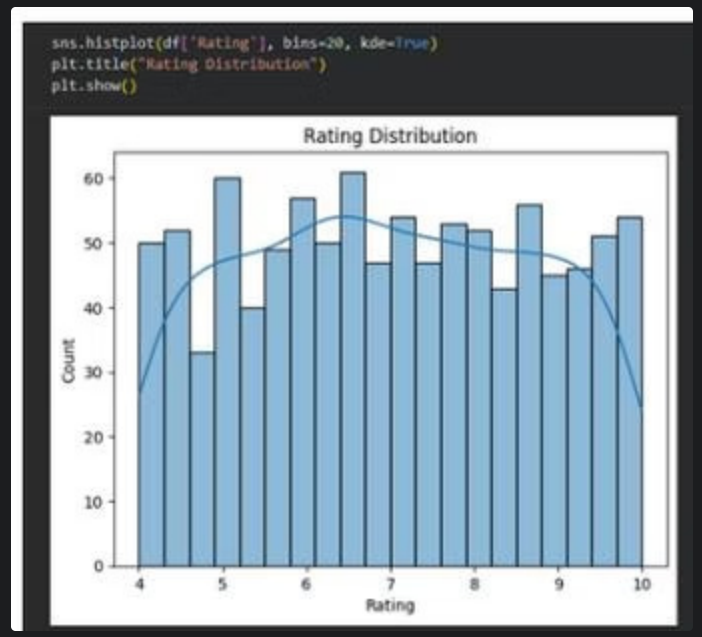
Product Line & Rating Distributions





Top Product Lines

Fashion accessories & Food and beverages lead with ~175 units each

Lowest Volume

Health and beauty trails at ~150 units

Rating Range

Roughly bell-shaped, centered around **6.5–7.0** on a 4–10 scale

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SQL Analysis: Sales & Customer Insights

Total Units Sold

5,510 units sold across all branches and product lines.

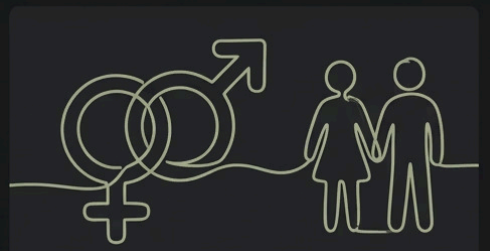
Gender Split

Nearly equal — **501 Female** vs. **499 Male** customers.

Average Unit Price

Mean of **$55.67**, ranging from $10.08 to $99.96.

SALES VOLUME: 5,510 TOTAL UNITS SOLD ACROSS ALL BRANCHES.



GENDER DISTRIBTION: 501 FEMALE, 499 MALE CUSTOMERS — NEARLY EQUAL SPLIT.

PAYMENT METHODS: DEBIT 345, CASH 311, CREDIT CARD — DEBIT IS MOST COMMON.

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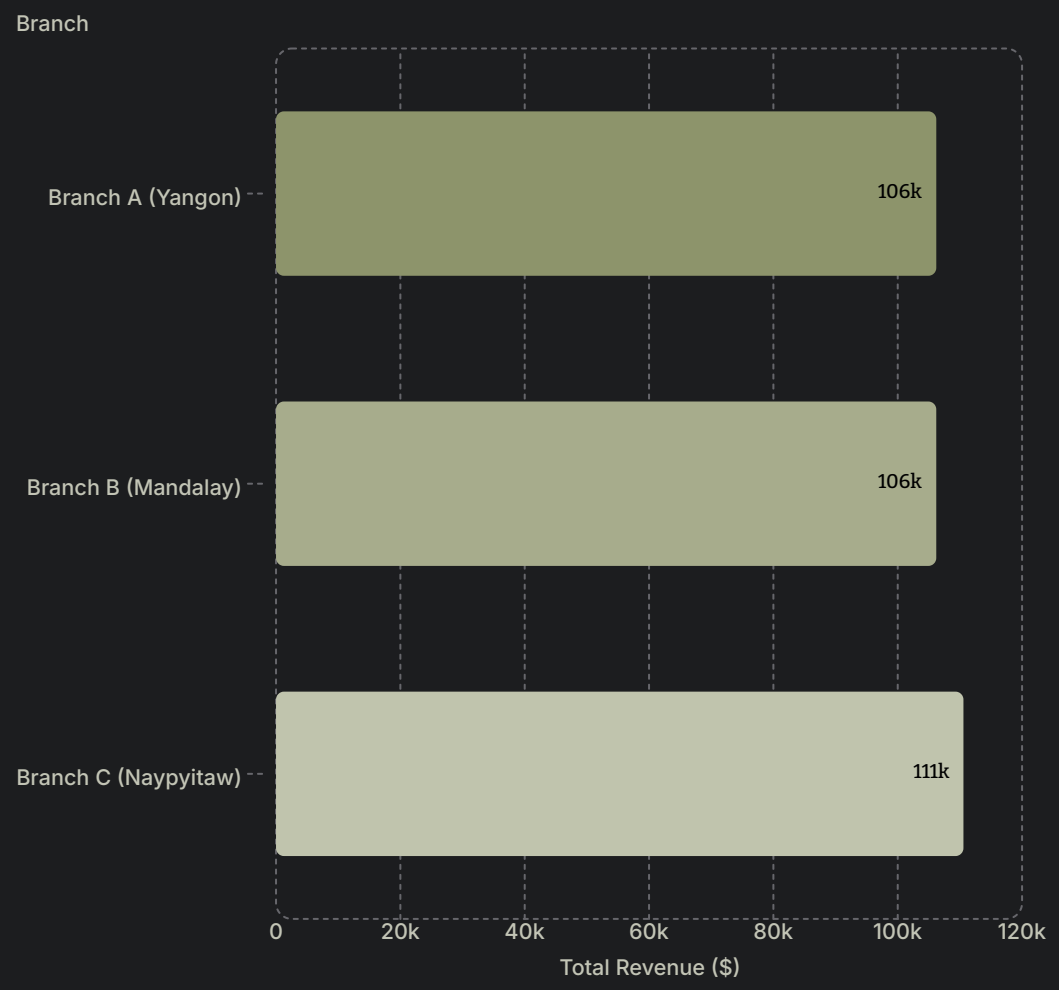
Branch Revenue & Ratings



Best Performing Branch

Branch C (Naypyitaw) leads on both revenue and customer satisfaction.

Branch C Revenue

$110,568.86 — highest of all three

Branch C Avg Rating

7.07 — highest average customer rating

Branch A Avg Rating

7.03 — close second

Branch B Avg Rating

6.82 — lowest rating among branches

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Preferred Payment Methods by City

Digital Payments Dominate

E-wallet is the top choice in both **Osaka** and **Kyoto**, while **Tokyo** customers prefer Cash.

Debit

345 transactions — overall most used

Cash

344 transactions

Credit Card

311 transactions

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Key Takeaways

Product Rating

Average product rating is around **7.5–8.0**, indicating generally positive customer satisfaction.

Branch C is the Clear Leader

Highest revenue ($110,568.86) and highest average rating (7.07) among all three branches.

Sales Are Right-Skewed

Most transactions are low-to-mid value; high-value purchases are rare but impactful.

Balanced Demographics

Near-equal gender split and evenly distributed product line volumes signal broad market appeal.

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